matheus hoffmann

the aquatic prometheus

sr. software engineer and jr. researcher, with 8 years of experience building scalable solutions, mainly for payment processing and fraud-fighting. my research topics are wave modelling, game theory and self-improving apps. i work in the open, and i'm open to commercial offers.

mhoffmannfs@gmail.com · linkedin · github · the site · ways of working

the arc

2026	**marola** and **Ouroboric** (BR): open research, with friends.
2025–2026	**ITV** (UK): consulting services, for one year.
2023–2024	**Signifyd** (US): senior software and ml engineer, fraud-fighting. in New York for the work.
2021–2022	**Elemeno AI** (US): lead mlops, for Americanas (BR).
2020	**Broad** (UK, YC W21): senior software engineer, a digital bank.
2017–2019	**Stone** (BR, NASDAQ): software engineer, payments. joined before the IPO.
2010–2020	**CEFET/RJ**, electronics technician; then **UFRJ**, with **LIOC**, its ocean lab.

the whole story, with the knowledge graph of the works, is on the site.

the works, each with a DOI

marola 10.5281/zenodo.23224155
a citizen-science ocean intelligence layer for the brazilian coast: "can i swim tomorrow, and when?". scala 3, rag with citations, designed in public (MIPs).

ww3-gpu 10.5281/zenodo.23221351
a kernel of NOAA's WAVEWATCH III wave model, ported to c++ and kokkos. it matches the fortran bit for bit on serial, openmp and cuda.

marola-bench 10.5281/zenodo.23352633
a proposal: a WeatherBench for brazil, scored against buoys.

picasso 10.5281/zenodo.23311208
my nix labs: flakes that pin every toolchain the other works build with.

closed source: the god that failed
10.5281/zenodo.23333656
a position paper on reverse engineering and the ethics of citation in the age of coding agents. not peer reviewed.

self-improving apps
writing and doing research for ouroboric-dev: open factories that improve themselves.

how i work

async and remote first, with 4h of overlap a day at most. no live coding: a code interview, or a talk about past projects. money first. references on request.

certs

AWS certified 4x: Cloud Practitioner, Machine Learning Specialty, Machine Learning Engineer Associate, Data Engineer Associate. **PADI certified 3x**: Advanced Open Water Diver, Enriched Air Nitrox Diver, Rescue Diver.

the sea

3rd place, Oceanman 10 km, rio de janeiro, september 2025. 2× the 10 km. strava.

every link in this pdf is checked on each pull request and each merge to main, by live links. the long form of this cv is cv.full.pdf. built from commit 9c61c7c.